**Rajalakshmi Engineering College**

Department of Computer Science and Design

## GE23627 DESIGN THINKING AND INNOVATION

## AY 2026-2027

# S.C.O.P.E.S – Team Scope Sheet

Scoping of design challenge When you want to work on real project using Design Thinking, begin the process with a specific and purposeful situation or problem to solve called Design Challenge. The Design Challenge should be doable, clear, actionable and hopefully inspiring. Your problem statement should be an outcome and not a solution.
You can use SCOPES as a tool to brainstorm and frame your problem.

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| **TEAM NAME** | |
| **TEAM MEMBERS** | **1.) AISHWARYA B [241701003]**<br>**2.) LATHA A [241701024]** |
| **DOMAIN NAME** | **IN-CAMPUS FOOD ORDERING SYSTEM** |
| **DATE** | **30/07/2026** |
| **FACULTY INCHARGE** | **Dr.P. Revathy** |

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| **S – Situation / Problem**<br>• What is the real situation you want to address?<br>• Where does this problem occur?<br>• Describe without mentioning any solution. | • Students face long kiosk queues to order and collect food, wasting valuable break time.<br>• No way to check food availability or crowd status at a canteen without physically going there.<br>• Orders are walk-in only, with no option to schedule a pickup time in advance.<br>• Canteen staff can't predict required food quantity, leading to shortages or wastage.<br>• No centralized system for admins to track orders and demand across multiple campus canteens. |

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| **C – Constraints**<br><br>What are the CONSTRAINTS when trying to address the situation or problem? | • Must work within limited canteen infrastructure (kitchen capacity, staff, and equipment can't be scaled up quickly).<br>• App needs to function reliably during peak hours with many simultaneous users without crashing or delaying orders.<br>• Solution must be simple enough for students, canteen staff, and admins with varying levels of tech familiarity to use easily.<br>• Limited time/budget typical of a student or campus-level project restricts the scope of features that can be built and tested. |
| **O – Objectives & Outcomes**<br><br>What are the OBJECTIVES and intended OUTCOME of this design challenge? | **Objectives**<br>• Reduce or eliminate physical queues at canteen kiosks by enabling pre-ordering through the app.<br>• Allow students/staff to select a convenient pickup time and view real-time food availability across canteens.<br>• Enable canteen admins to track order volume and plan food quantity accurately, minimizing shortages and wastage.<br>• Provide role-based access (student/staff, admin) for a smooth, organized ordering and management experience.<br><br>**Intended Outcome**<br>• Students and staff save time by skipping queues and picking up food exactly when it's ready.<br>• Canteens experience better-organized order flow, especially during peak hours.<br>• Admins get clear data on demand patterns, enabling efficient food preparation and reduced wastage.<br>• Overall improvement in campus food service efficiency, convenience, and satisfaction for all stakeholders. |
| **P – People**<br><br>Who are the PEOPLE that you are trying to help? Whose problem are you trying to address?<br><br>Primary Stakeholders | **<u>Primary Stakeholders</u>**<br><br>• **Students** who order food during breaks and want to save time by avoiding queues.<br><br>• **College staff/faculty** who need quick, convenient food ordering during limited free time. |

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| Secondary Stakeholders<br><br>Decision Makers | **<u>Secondary Stakeholders</u>**<br>• **Canteen owners/vendors** and kitchen staff who prepare and manage food orders.<br>**<u>Decision Makers</u>**<br>• **College administration/management** who approve and oversee implementation of the app on campus.<br>• App admin(s) who manage canteen listings, monitor orders, and control overall platform operations. |
| **E – Estimates**<br>What are the ESTIMATED resources required for this design challenge? | Figma, Google Forms, Canva/Adobe Illustrator laptop, smartphone, internet, user interviews, surveys, literature review, case studies, paper/mobile prototypes, and usability testing with feedback documentation. |
| **S – Scope to Explore**<br>Is there ample SCOPE to explore multiple alternatives to address the problem? | • Different ordering models: pure pre-order vs. hybrid (walk-in + pre-order).<br>• Queue handling options: live wait-time display, token system, or slot-based pickup.<br>• Integration approach: single unified app for all canteens vs. separate dashboards linked to central admin.<br>• Admin tools can range from basic order tracking to advanced demand forecasting. |

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| **Frame the Design Challenge** | **How might we design a campus food ordering app that allows students and staff to pre-order food, select a convenient pickup time, and avoid long kiosk queues**<br><br>**while helping canteen admins efficiently manage food quantity, orders, and demand across multiple canteens on campus?** |

**SIGNATURE**